

PART 1 Crop Yield

I tested p values and considered dropping the Special variable but left it because it has a very small p value much smaller than 0.5, as did the other predictors. It's reasonable that a crop with or without special treatment could yield better results.

I did not run multinomial logistic regression since there are only two classes in the outcome variable.

I only ran Support Vector machine with the 'linear' kernel for sake of time but could have tested with rbf, poly, and sigmoid

Model	Accuracy	Precision (0, 1)	Recall (0, 1)
Logistic Regression			
Original	0.77	0.66, 0.77	0.17, 0.97
Oversampled	0.71	0.46, 0.90	0.77, 0.69
Undersampled	0.70	0.45, 0.90	0.77, 0.68*
Classification Trees			
Original	0.71	0.41, 0.79	0.36, 0.82
Oversampled	0.67	0.38, 0.80	0.46, 0.75
Undersampled	0.62	0.36, 0.83	0.64, 0.61*
Pruned Trees			
Original	0.73	0.46, 0.78	0.28, 0.89
Oversampled	0.65	0.38, 0.84	0.62, 0.65*
Undersampled	0.63	0.38, 0.86	0.70, 0.61
Gaussian Naïve Bayes			
Original	0.77	0.59, 0.80	0.32, 0.93
Oversampled	0.62	0.38, 0.89	0.80, 0.56
Undersampled	0.63	0.38, 0.89	0.79, 0.57
KNN k=3 (also tried 1, 5, 7, 9, 11, and 13)			
Original	0.73	0.46, 0.79	0.29, 0.89
Oversampled	0.64	0.37, 0.83	0.61, 0.65*
Undersampled	0.60	0.35, 0.85	0.70, 0.57
Bagging			
Original	0.75	0.00, 0.75	0.00, 1.00
Oversampled	0.59	0.36, 0.87	0.77, 0.53
Undersampled	0.67	0.41, 0.87	0.70, 0.65*
Boosting			
Original	0.76	0.57, 0.77	0.13, 0.97
Oversampled	0.71	0.45, 0.87	0.68, 0.72
Undersampled	0.67	0.41, 0.86	0.69, 0.66*
Random Forest			

Original	0.76	0.58, 0.79	0.24, 0.94
Oversampled	0.73	0.48, 0.85	0.57, 0.79
Undersampled	0.67	0.42, 0.88	0.74, 0.65
Neural Net			
Original	0.78	0.70, 0.79	0.23, 0.97
Oversampled	0.62	0.39, 0.91	0.83, 0.55
Undersampled	0.63	0.40, 0.92	0.86, 0.56
Discriminant Analysis			
Original	0.77	0.71, 0.77	0.16, 0.98
Oversampled	0.70	0.45, 0.90	0.79, 0.67
Undersampled	0.70	0.45, 0.90	0.77, 0.68
Support Vector Machine			
Original	0.75	0.00, 0.75	0.00, 1.00
Oversampled	0.69	0.44, 0.90	0.79, 0.66
Undersampled	0.69	0.44, 0.91	0.80, 0.66

Recall balances of top performing models (max-min):

Logistic US = 0.09

Classification Tree US = 0.03

Pruned Tree OS = 0.03

KNN OS (k=3) = 0.04

Bagging US = 0.05

Boosting US = 0.03

Precision averages of top performing models:

Logistic US = 0.675

Classification Tree US = 0.595

Pruned Tree OS = 0.61

KNN OS (k=3) = 0.6

Bagging US = 0.64

Boosting US = 0.635

The bagging undersampled and boosting undersampled models both appear to perform the best and neither appears to be overfit. The bagging US model has slightly better precisions but the recall balance of the boosting US is significantly better. **Therefore, I determine the boosting under sampled model to be the best model.**

PART 2 Glass.CSV

I chose to drop the RI variable, but could have left it in first. However, it did have a much higher p value than the other predictors (0.249).

Models I did not run for sake of time but could have include: Random Forest, Discriminant Analysis, Neural Network, Support Vector Machine

Model	Accuracy	Precision (1, 2, 3, 4, 5, 6, 7)	Recall (1, 2, 3, 4, 5, 6, 7)
KNN k =9 (also tried, 1,3,5,7 for k)			
Original	0.65	0.53, 0.56, 0.00, 1.00, 1.00, 1.00	0.73, 0.71, 0.00, 0.25, 0.67, 0.88
Oversampled	0.72	0.67, 0.82, 0.29, 0.80, 0.75, 1.00	0.55, 0.64, 0.67, 1.00, 1.00, 0.88*
Undersampled	0.60	0.71, 0.62, 0.40, 0.67, 0.38, 1.00	0.45, 0.71, 0.67, 0.50, 1.00, 0.50
Classification Trees			
Original	0.81	0.64, 0.92, 0.50, 1.00, 1.00, 0.89	0.82, 0.79, 0.33, 1.00, 0.67, 1.00
Oversampled	0.79	0.78, 0.90, 0.50, 0.67, 1.00, 0.89	0.64, 0.64, 1.00, 1.00, 1.00, 1.00*
Undersampled	0.63	0.83, 0.67, 0.25, 0.75, 0.75, 0.67	0.45, 0.43, 0.67, 0.75, 1.00, 1.00
Pruned Trees			
Original	0.77	0.56, 0.82, 0.50, 1.00, 1.00, 1.00	0.82, 0.64, 0.33, 0.75, 1.00, 1.00
Oversampled	0.79	1.00, 0.79, 0.40, 0.67, 1.00, 0.89	0.55, 0.79, 0.67, 1.00, 1.00, 1.00*
Undersampled	0.67	0.67, 0.71, 0.33, 0.75, 0.50, 0.78	0.55, 0.71, 0.33, 0.75, 0.67, 0.88
Gaussian Naïve Bayes			
Original	0.53	0.39, 0.43, 0.25, 0.50, 1.00, 0.89	0.64, 0.21, 0.33, 0.25, 1.00, 1.00
Oversampled	0.49	0.29, 0.60, 0.20, 0.75, 1.00, 0.80	0.18, 0.21, 1.00, 0.75, 0.67, 1.00

Undersampled	0.33	0.75, 0.50, 0.12, 0.21, 1.00, 0.75	0.27, 0.07, 0.67, 0.75, 0.67, 0.38
Bagging			
Original	0.81	0.67, 0.83, 0.00, 0.80, 1.00, 1.00	0.91, 0.71, 0.00, 1.00, 1.00, 1.00
Oversampled	0.77	0.64, 0.90, 0.33, 0.80, 1.00, 1.00	0.64, 0.64, 0.67, 1.00, 1.00, 1.00*
Undersampled	0.56	0.67, 0.50, 0.20, 0.80, 0.60, 0.88	0.18, 0.43, 0.67, 1.00, 1.00, 0.88
Boosting			
Original	0.79	0.71, 0.77, 1.00, 0.67, 1.00, 0.89	0.91, 0.71, 0.33, 0.50, 1.00, 1.00
Oversampled	0.81	0.80, 0.83, 0.60, 0.75, 1.00, 0.89	0.73, 0.71, 1.00, 0.75, 1.00, 1.00*
Undersampled	0.60	1.00, 0.50, 0.30, 0.67, 0.75, 0.89	0.18, 0.43, 1.00, 1.00, 1.00, 1.00
Multinomial Logistic Regression			
Original	0.70	0.75, 0.60, 0.00, 1.00, 0.00, 0.78	0.82, 0.86, 0.00, 0.50, 0.00, 0.88
Oversampled	0.65	0.70, 0.80, 0.20, 0.67, 1.00, 0.89	0.64, 0.29, 0.67, 1.00, 1.00, 1.00
Undersampled	0.65	0.50, 0.59, 0.17, 0.80, 1.00, 1.00	0.18, 0.71, 0.33, 1.00, 1.00, 1.00

Recall balances of top performing models (max-min):

KNN OS (k=9) = 0.45

Classification Tree OS = 0.36

Pruned Tree OS = 0.45

Bagging OS = 0.36

Boosting OS = 0.29

Precision averages of top performing models:

KNN OS (k=9) = 0.72

Classification Tree OS = 0.79

Pruned Tree OS = 0.79

Bagging OS = 0.78

Boosting OS = 0.81

The bagging OS, boosting OS, and Classification Tree OS have the best recalls and all have fairly good precisions. There appears to be some overfitting among precision and recall values in all models. **Despite close options, the Boosting OS model has the best recall balance and best precisions, therefore it's the best model.**